Understanding Customer Feedback on Food Products

Sentiment classification and exploratory themes in Amazon Fine Food Reviews

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Executive overview

The supplied 568,454 reviews were cleaned and reduced to 392,868 unique usable normalised texts. A sentiment-stratified sample of 50,000 supports a 40,000/10,000 training/test evaluation. The selected tuned classifier achieves 77.9% test accuracy and macro-F1 0.614. Six exploratory topics describe recurring product discussions. The analysis identifies review patterns for human investigation; ratings, historical coverage and modelling limitations constrain conclusions.

Contents and decision summary

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What the client can use

The sentiment model can help organise reviews for review by staff. Topic summaries show what people discuss and the balance of ratings within those discussions. They do not prove why customers behave as they do, identify every individual concern, or measure current satisfaction.

Use negative-review volume and share together when prioritising investigations. Keep human review for ambiguous or neutral feedback, and validate the approach on recent reviews before operational use.

1. Business objectives and dataset

1.1 Questions addressed

The retailer needs a structured view of customer praise and dissatisfaction. This analysis asks which themes recur, whether review text predicts rating-derived sentiment, which classes are confused, and whether a controlled model change improves class-balanced performance. Two separate workflows answer these questions: supervised sentiment classification and unsupervised LDA theme discovery.

1.2 Source and suitability

The CSV is the Amazon Fine Food Reviews distribution on Kaggle (SNAP, n.d.). The original Stanford source describes 568,454 fine-food reviews extending up to October 2012, including text, ratings, product and user information. It supports historical text analysis, not present-day market inference (Stanford Network Analysis Project, n.d.).

Dataset item	Observed value
Rows / columns	568,454 / 10
Input to classifier	Full review text after fixed cleaning
Negative label	1–2 stars
Neutral label	3 stars
Positive label	4–5 stars
Missing text / ratings	0 / 0
Missing profile names / summaries	26 / 27
Repeated raw texts after first	174,875

1.3 Ethical and analytical boundaries

Names and reviewer identifiers are excluded from model inputs and report examples. Ratings are noisy sentiment proxies: a three-star score need not correspond to linguistically neutral wording. Self-selection, product mix and historical scope limit generalization. Positive reviews dominate, so minority-class performance must accompany accuracy.

2. Preparation and sample

The raw data frame is preserved. Invalid ratings and unusable texts are filtered. HTML entities are decoded, tags removed and casing standardized. Negative contractions are expanded before punctuation, and digits are replaced by spaces. Common stop words are removed while no, not, nor and never are retained. No lemmatization is performed. Removing words can still alter phrase context.

Cleaning stage	Records
Original reviews	568,454
Invalid text/rating removed	0
Empty after cleaning	8
Conflicting clean-text sentiment records removed	215
Further repeated clean texts removed	175,363
Retained reviews	392,868

All occurrences of a normalized text linked to more than one sentiment category are excluded; one record is retained per other normalized text. Differences in star rating within the same sentiment category do not trigger conflict removal. This reduces exact input duplication but can discard legitimate repeated feedback. The retained population differs from all original submissions.

Retained sentiment	Reviews	Share
negative	56,925	14.49%
neutral	29,682	7.56%
positive	306,261	77.96%

A reproducible stratified sample of 50,000 reviews is selected, then divided 80:20 into 40,000 training and 10,000 test reviews, with seed 42. An assertion checks that identical cleaned inputs do not appear in both. Sampling preserves sentiment proportions approximately, not necessarily every product, year or writing style. Vocabulary learning and tuning use training data only.

3. Exploratory findings

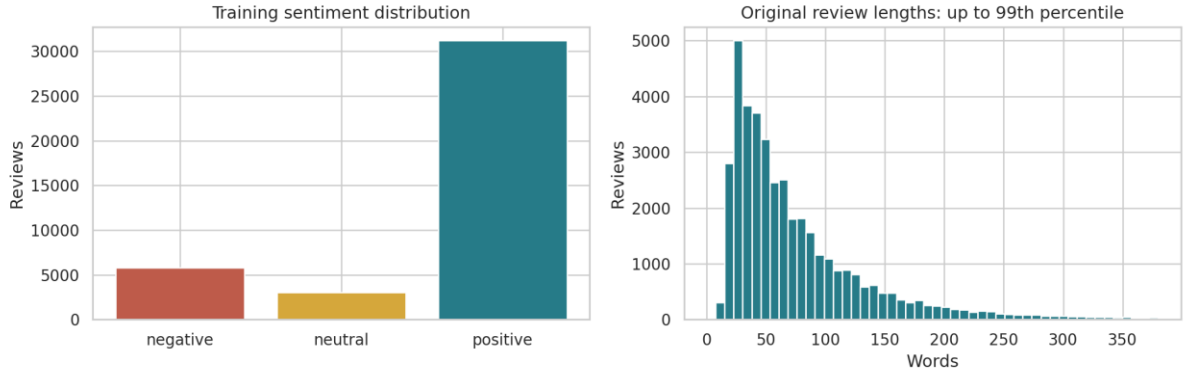


Figure 1. Training sentiment counts and original-text review lengths.

The retained data remain approximately 78% positive, 14.5% negative and 7.6% neutral. Training review length has median 57 words and mean 79.8; the 99th percentile is 380. The histogram restricts its display range for readability but removes no long reviews from modelling.



Figure 2. Word frequency within each rating-derived sentiment class.

The most frequent training terms include not, like, good, just, taste, great, product, coffee. Frequencies and word clouds show discussion volume, not whether a particular product characteristic caused praise or criticism. Comparing clouds is qualitative because class sizes differ; topic sentiment tables provide explicit denominators.

Historical tables cover years with at least 50 sampled training reviews (2005–2012). Counts are saved alongside shares. These are historical rating patterns, not current satisfaction or a forecasting evaluation; deduplication can alter the time distribution.

4. Features and model development

4.1 Two feature representations

TF-IDF converts text to sparse numerical weights for sentiment classification. Count-based features supply non-negative occurrences for LDA. Unigrams represent individual terms; bigrams add two-word sequences but do not restore full context after stopword removal. Vocabulary limits control computation rather than proving all omitted terms unimportant (scikit-learn developers, n.d.b).

4.2 Pipeline and baseline

The baseline combines TfidfVectorizer(max_features=10000, min_df=2, ngram_range=(1,1)) and logistic regression with C=1.0, L2 regularisation, lbfgs and mixite=1000. It is an efficient sparse-text baseline, with limitations on sarcasm and contextual language. A pipeline fits vocabulary and IDF within each training fold to prevent leakage (scikit-learn developers, n.d.a).

4.3 Controlled improvement experiment

Search component	Candidates
N-grams	Unigrams; unigrams + bigrams
Regularisation C	0.5; 2.0
Class weighting	None; balanced
Validation	3 stratified folds, seed 42
Selection metric	Macro-F1
Search size	8 configurations × 3 folds

Baseline validation macro-F1 is 0.5644 (SD 0.0068). The best grid score is 0.6159. Winning settings are ngram_range=[1, 2], C=0.5, class_weight=balanced. The selected model is tuned, designated from training-validation scores before the test comparison. Balanced weighting increases minority-class training contribution; larger C weakens regularisation (scikit-learn developers, n.d.c).

The test set is withheld from fitting and tuning. Winning cross-validation scores are not an unbiased nested estimate of the full search. Ratings create targets only and are never numerical predictors.

5. Sentiment evaluation

Model	Test accuracy	Macro-F1	Weighted-F1
Majority	77.95%	0.2920	0.6829
Baseline	84.91%	0.5721	0.8224
Tuned	77.92%	0.6141	0.8058

Selected-model macro-F1 changes by +0.0420 and accuracy by -6.99 percentage points relative to the baseline. Improvement is judged against the chosen class-balanced criterion, not a promise to raise every metric. Macro-F1 gives each class equal weight; weighted-F1 gives frequent classes greater influence (scikit-learn developers, n.d.d).

Selected-model class	Precision	Recall	F1	Support
negative	0.586	0.713	0.643	1449
neutral	0.234	0.483	0.315	756
positive	0.957	0.820	0.884	7795

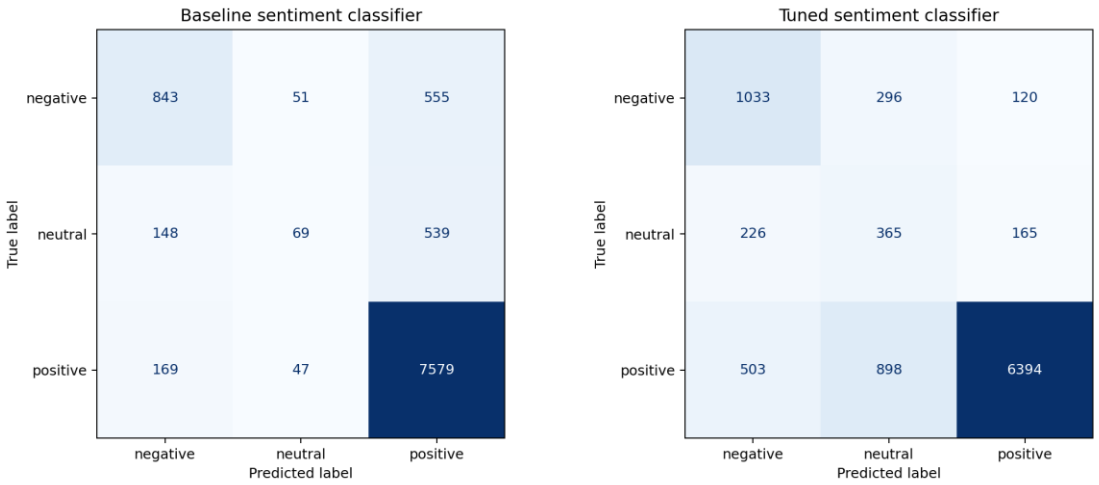


Figure 3. Actual sentiment is in rows; predictions are in columns.

Neutral recall is 48.3%. Ambiguous three-star wording and noisy labels may contribute to errors; these are hypotheses from review inspection rather than established causes. Accuracy alone obscures this weakness. The notebook includes misclassified text excerpts without reviewer identities. Test results do not change the selected configuration.

6. Exploratory themes

Six-topic online LDA is fitted to training word counts with max_features=5000, min_df=5, max_df=0.95, ten iterations and seed 42. Each review has a topic mixture; the strongest topic is used for summary counts. Six is an exploratory starting point, not an optimised topic count (scikit-learn developers, n.d.e).

Topic / tentative label	Key terms	Reviews	Negative share
1. Taste and food preferences	not, like, taste, good, flavor, just, chocolate	13,504	13.2%
2. Pet food and treat acceptance	not, food, dog, did, like, product, old	5,721	23.9%
3. Tea and other drinks	tea, not, drink, like, taste, water, flavor	4,662	13.0%
4. Preparation and product use	use, not, make, product, used, water, chicken	3,427	12.9%
5. Purchasing and value	amazon, great, price, product, love, store, good	7,896	10.8%
6. Coffee flavour and preparation	coffee, not, cup, like, flavor, good, taste	4,790	15.6%



Figure 4. Dominant-topic volume and rating-derived sentiment composition.

Labels were assigned after examining keywords and two topic-specific training excerpts per topic. Some high-weight excerpts are repetitive and are not treated as substantive label evidence. This small interpretive check is not comprehensive validation. The notebook retains excerpts and the top 12 keywords. Generic and overlapping vocabulary limits distinctiveness. Classification F1 does not apply to unlabelled topics.

7. Actions, limitations and conclusion

Review theme	Suggested investigation
Taste and food preferences	Review taste and sweetness complaints by product before changing recipes; preserve flavour characteristics that customers praise.
Pet food and treat acceptance	Prioritise a product-level review of pet-food and treat acceptance complaints; check the volume and wording before drawing conclusions.
Tea and other drinks	Check drink flavour and sweetness descriptions against negative feedback; inspect mixed-topic reviews manually.
Preparation and product use	Check preparation and storage instructions and packaging usability against additional reviews before changing product guidance.
Purchasing and value	Preserve well-received value offers and investigate ordering complaints separately; verify issues against transaction evidence.
Coffee flavour and preparation	Review coffee flavour and strength feedback by product and preparation method; manually screen repetitive text before using it as evidence.

Prioritise using both negative share and negative-review count; high rates in small groups can be misleading. These signals guide manual investigation, not product-quality verdicts or financial-loss estimates. Preserve commonly praised features while checking complaints against additional reviews and operational evidence.

Limitations

Historical coverage, self-selection and sentiment-stratified sampling limit representativeness. Ratings are proxies, not independently annotated emotions. Cleaning and conflict exclusion alter the population; conflict detection is corpus-wide before splitting. Exact input overlap is prevented, but near duplicates and shared products/reviewers may cross the random split. Grouped and chronological evaluation would be stronger checks for new-product or future-review deployment.

The fixed stopword rules can alter phrasing. No lemmatisation, calibration assessment, nested search evaluation or external validation is performed. LDA labels are tentative, with only one topic count and seed tested; no coherence/stability study establishes optimal themes. The per-class metrics and examples expose these limits rather than treating high aggregate accuracy as universal success.

Conclusion

The notebook completes cleaning, exploration analysis, sentiment pipelines, controlled model comparison and six-topic discovery. The selected classifier obtains test macro-F1 0.614, with important class-level trade-offs. The outputs provide a structured basis for human review. Validate with recent, manually labelled data and monitor minority-class errors before using automatic alerts.

8. References

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